YaniMed
System Design & Architecture – Revision 2

AI-powered PDF-grounded medical reviewer with integrated Mock Test module

DOCUMENT STATUS	SCOPE
Revision 2 of the proposed blueprint. Supersedes v1. Still a design: implementation requires repository verification (milestone M1).	Shared Subjects and PDFs, ingestion and retrieval, question generation and validation, Mock Test exam flow, security, database constraints, operations, testing and delivery phases.

Deployment assumption

YaniMed is treated as single-user and localhost-run. v2 sizes controls for that (one in-process worker, no per-user quotas) but keeps `userId` on every owned row so going multi-user needs no schema change. Items that change if you host it publicly are tagged **[PUBLIC]**. Confirm this in M0.

Design principles

- One shared Subject entity across Subjects and Mock Test; no duplicate subject table, no duplicate PDF objects.
- Only uploaded and successfully processed PDFs ground medical answers and exam questions. No web fallback.
- Exactly 50 validated questions per playable attempt. Partial sets are never published.
- **Constraints before code paths.** Every rule that can be a database constraint is one. Application logic is the second line of defense, not the first.
- **No model output is trusted.** It is checked structurally, against the source text, and by an independent verifier pass.
- Durable jobs: state lives in Postgres, never only in a request, a tab, or process memory.
- **Smallest safe v1.** Offline sync and explicit session takeover are deferred; everything else ships tested.
- Preserve existing features: additive migrations, feature flag, per-milestone regression.

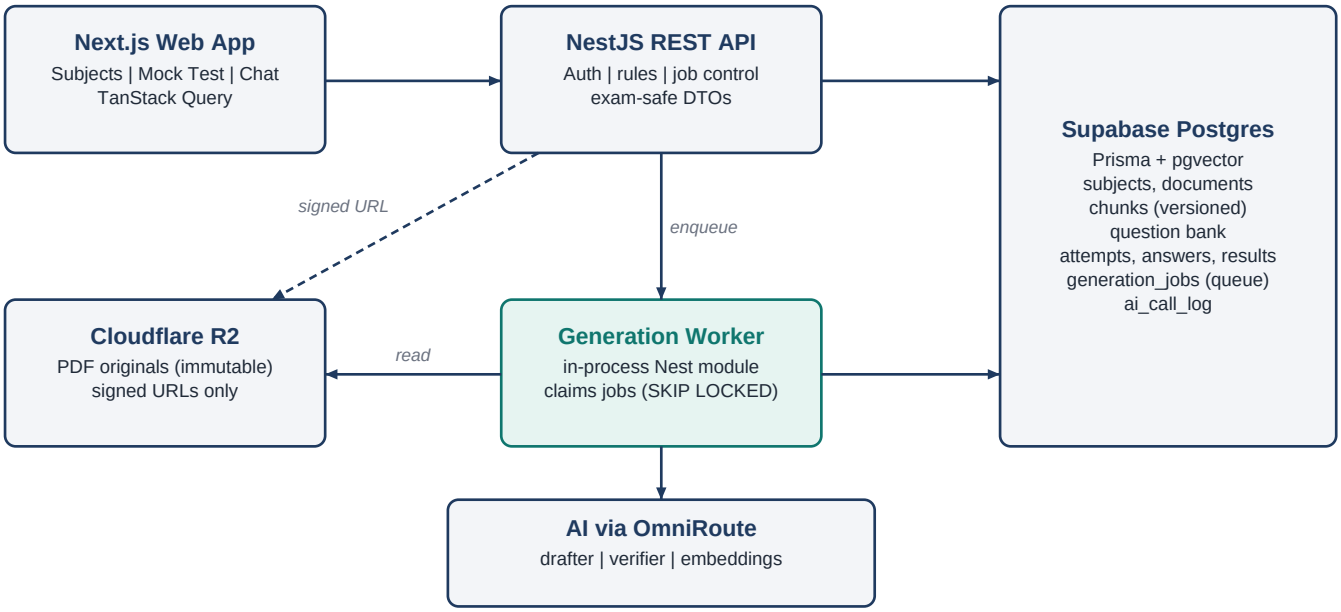
What changed from v1

#	Gap in v1	Fix in v2	Section
1	Validation was named but not defined	Layered validation: deterministic rules, independent verifier, citation quote match, embedding duplicate check, full-set check	6
2	No coverage strategy	Per-subject topic list, slot planning, per-topic cap, balanced answer positions	6
3	Undefined numbers ("sufficient", "bounded")	Concrete defaults for every limit, kept in config	App. A
4	Queue technology undecided	Postgres job table with row-lock claim; in-process worker; boot recovery	1, 6, 10
5	Content changing mid-flight	Document snapshot per job, versioned chunks, embedding model tags	4, 6
6	Stuck states	ABANDONED + Discard, watchdog, partial unique indexes	5, 7
7	Retry semantics ambiguous	Per-subject question bank; Retry samples unseen questions first	2, 8
8	Reports went nowhere	Report lifecycle with automatic quarantine	8
9	No cost or abuse controls	Job caps, daily AI budget, circuit breaker, call log	10
10	Upload and lifecycle hygiene	Limits, magic-byte checks, OCR cap, reconciliation, backups, disclaimer	4, 10
11	No traceability or AI testing	Provenance on every question, golden set, structured output only	5, 11

#	Gap in v1	Fix in v2	Section
12	Offline sync and takeover risk	Deferred to v1.1; online-only autosave, latest session wins	7, 13
13	Plan front-loaded the safe work	M0 decisions, M2 generation spike, tests per milestone, feature flag	12

1. Executive architecture

YaniMed is a PDF-grounded study application. Users organize reviewer PDFs by subject, ask citation-backed questions, and take subject-specific mock tests. The web client talks to a NestJS backend that owns authorization, persistence, retrieval, generation, validation and grading. Long-running generation runs in a worker module that is driven by a job table in Postgres, so no HTTP request or browser tab is ever the execution context.



Proposed logical architecture – exact integrations require repository verification

Core technology decisions

Layer	Planned technology / responsibility
Client	Next.js, TypeScript, Tailwind CSS, shadcn/ui; TanStack Query and Zustand as appropriate.
API	NestJS, whitelisted and validated DTOs, REST, stable machine-readable error codes (section 9).
Job runner	Worker module inside the NestJS process. Claims rows from `generation_jobs` with `SELECT ... FOR UPDATE SKIP LOCKED`. No Redis, no new infrastructure. **[PUBLIC]** run the same module as a separate process or service.
Database	Supabase PostgreSQL, Prisma, pgvector. Partial unique indexes, CHECK constraints and triggers are added as reviewed raw SQL inside Prisma migrations, because the Prisma schema cannot express them.
File storage	Cloudflare R2 for original PDFs. Objects are immutable once written. Access only through short-lived signed URLs issued after an ownership check.
PDF ingestion	Text extraction by physical page; OCR for pages with inadequate text; per-document timeout and OCR cap; versioned index swap.
AI	Gemini embeddings (model id and dimension stored on every chunk). OmniRoute for drafting and verification, with the verifier on a different model or combo when the gateway offers one.
Observability	Structured logs carrying jobId and attemptId, an `ai_call_log` table, and a health endpoint for the Mock Test subsystem.

2. Phase 0 decision log

These were open in v1 and every later phase depends on them. Each row is a proposed default; confirm or change it in milestone M0, before schema work starts.

ID	Decision	Default	Why
D1	Retry semantics	Per-subject **question bank**. Retry builds a new attempt from 50 bank questions, unseen first, then least recently seen. Generation runs only when too few eligible questions exist, or when the user asks for new ones.	Cuts AI cost, keeps questions immutable, and makes report quarantine possible.
D2	Queue	Postgres table with row-lock claim.	No new infrastructure; state survives restarts.
D3	Worker hosting	In-process with the API. Stops when the API stops; jobs resume on boot.	Fits localhost. A sleeping host would otherwise break recovery. **[PUBLIC]** separate process.
D4	Offline	v1 online-only. Banner and disabled controls when offline. Offline sync is v1.1.	Largest source of subtle bugs, smallest launch value.
D5	Session ownership	Latest session wins via `sessionVersion`. Older tabs get 409 and read-only mode.	No lease, heartbeat or TTL needed in v1.
D6	Minimum content	At least 60 eligible chunks and 40 distinct pages (App. A).	Below this, 50 varied questions are not possible.
D7	Verifier	Independent pass; must agree with the key. Different model if available.	A model grading its own work approves its own mistakes.
D8	Topics	Controlled per-subject topic list, not free text from the model.	Free text fragments topic analytics ("Cardiology" vs "Cardiovascular").
D9	Unfinished attempts	Discard allowed any time; no auto-expiry; dashboard shows age.	Prevents permanent lock-out by an old attempt.
D10	Scores vs retired questions	Scores never change automatically. History shows a notice.	Matches the immutable-result rule.
D11	Spend control	One active job; per-job call caps; configurable daily AI-call budget.	Single-user needs a budget, not per-user quotas.

3. Shared Subjects architecture

Subjects and Mock Test are two views of the same domain data. Creating Pediatrics in Mock Test creates one subject record that also appears in Subjects. Attaching two PDFs shows "2 PDFs" on the Subjects page, never "50 items".

Display and synchronization rules

- Subjects page: every owned subject, including those with zero PDFs; name and PDF count only.
- Mock Test dashboard: subjects with at least one attached PDF. Generation additionally requires the content threshold (D6) and says exactly what is missing when it is not met.
- Create, rename, attach, delete and count changes use the same subject ID and attachment records. Relevant client queries are invalidated and refetched after every mutation.
- No PDF object is ever duplicated for Mock Test.

Deletion is a state, not an instant

- **Subject delete** (with confirmation): mark the subject DELETING (hidden everywhere) → cancel its active job → delete R2 objects → hard-delete rows in one transaction. If any step fails the row stays DELETING and the reconciler retries. Nothing is half-deleted and visible.
- **Single PDF delete**: mark the document DELETING and remove it from retrieval immediately; delete the R2 object; keep citation metadata, which then displays "Source Unavailable".
- A bank question whose citations are *all* unavailable becomes SOURCE_UNAVAILABLE: excluded from new attempts, kept in history. A question that still has at least one available citation stays ACTIVE.

4. PDF ingestion and retrieval

PDFs are uploaded once and processed into page-aware searchable content. Chat and Mock Test share the index but use different output policies.

Stage	Expected behavior	On failure
Upload	Authorize ownership; check magic bytes (`%PDF-`), MIME, size and page limits; reject encrypted PDFs with a clear message; store original in R2; create document record; flag exact content-hash duplicates within the subject.	Reject with a reason code. No orphan R2 object.
Extract	Text by actual physical page. OCR pages with inadequate text, up to the OCR cap. Run inside a per-document timeout.	FAILED with OCR_QUALITY if over 30% of pages are unreadable.
Chunk	Store document ID, physical page range, optional verified printed page and chapter. Mark `eligibleForQuestions=false` for front matter, contents, index, references, repeated headers and very short chunks (still searchable by chat).	Zero eligible chunks marks the document READY with a warning, not usable for Mock Test.
Embed / index	Embed with retry and rate-limit handling. Write chunks under a new `indexVersion`.	Resumable per chunk batch; FAILED after bounded retries.
Ready	In one transaction after integrity checks (chunks exist, every chunk has an embedding of the current dimension, page numbers within range), switch `activeIndexVersion`.	Old index stays active.
Retrieve	Scope strictly to the current user, selected subject, READY documents and their active index version. Never use web fallback.	Empty retrieval means "insufficient", never a guess.

Document status model

UPLOADED → EXTRACTING → (OCR) → EMBEDDING → READY. Side states: FAILED (with reason code), STALE_INDEX (embedding model changed), DELETING. Every status is visible to the user with a plain-language message.

Content that changes over time

- **Reprocessing** builds a new index version beside the old one and swaps atomically, so retrieval never sees a half-built document. Old versions are garbage-collected only when no running job is pinned to them.
- **Embedding model change**: chunks store `embeddingModel`. Retrieval queries only chunks matching the configured model; a change marks documents STALE_INDEX and queues re-embedding.
- **Immutable originals**: "replacing" a PDF means uploading a new document and deleting the old one. Citations therefore never point at a file that changed under them.

> **Citation invariant**
> Every cited question references a real source document and an actual physical PDF page, and the quoted supporting text is found on that page by a normalized match. Printed book pages and chapters are optional unless reliably verified. Deleted individual PDFs retain historical citation metadata and display "Source Unavailable".

Hygiene and reconciliation

- Reconciler (on demand and weekly): R2 keys with no database row older than 24 hours are removed after a dry-run report; database documents whose object is missing become FAILED(MISSING_OBJECT).
- Same-hash duplicate uploads warn instead of silently doubling OCR and embedding cost.

5. Mock Test domain model

Conceptual schema, not a mandate to duplicate models already in Prisma. The key change from v1: generated questions live in a **per-subject bank** and attempts reference them through an ordered join table.

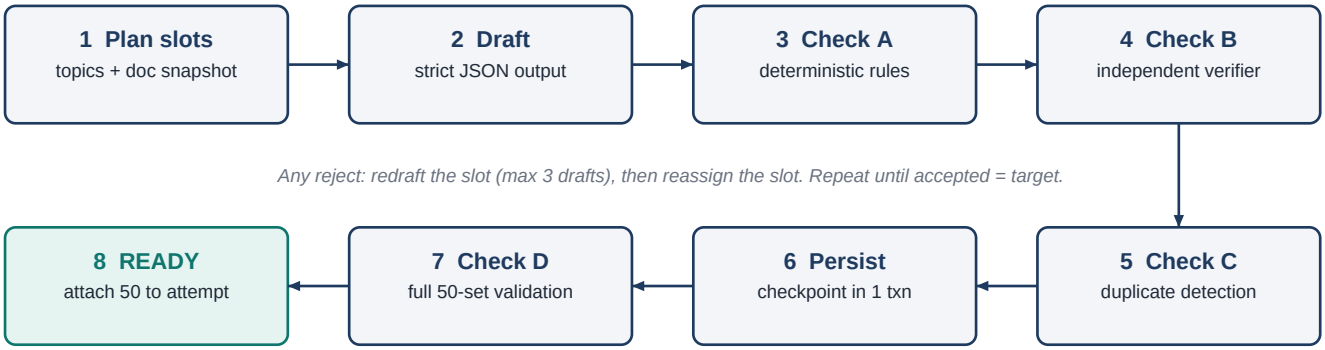
Entity	Purpose / key fields
Subject, Document, Chunk (existing, extended)	Subject status (ACTIVE/DELETING). Document status, failure reason, content hash, page count, activeIndexVersion. Chunk indexVersion, embeddingModel, physical page range, eligibleForQuestions.
SubjectTopic	Controlled topic list per subject; stable keys; new topics are appended, never renamed under existing questions.
GenerationJob	Subject, requested count, state, accepted count (0-50), plan JSON (slots), heartbeatAt, recovery count, AI-call counters, failure reason. **GenerationJobDocument** snapshots the document IDs and index versions used.

Entity	Purpose / key fields
MockQuestion (bank)	Subject, topic, vignette, lead-in, educational objective, status (ACTIVE / QUARANTINED / RETIRED / SOURCE_UNAVAILABLE). Provenance: job, draft model, verifier model, prompt version, validator version, verifier result. Content columns immutable.
MockChoice	Question, opaque ID, position 1-4, text, isCorrect, explanation (correct and each distractor).
QuestionCitation	Question, document ID (kept as snapshot title if the document is deleted), physical PDF page, optional verified printed page/chapter, supporting quote.
MockAttempt	User, subject, status, sessionVersion, cursorPosition, lastActivityAt, optional job link.
AttemptQuestion	Attempt, question, position 1-50, persisted choice order, status UNSEEN/SKIPPED/ANSWERED. Order is generated once and never reshuffled.
AttemptAnswer	One per attempt question: choice, server-assigned revision, idempotency key.
AttemptResult	Immutable score snapshot: per-question selected, correct and topic; per-topic performance.
QuestionReport	Reporter, attempt, question, reason, details, status OPEN/ACCEPTED/DISMISSED, resolution note.
AiCallLog	Job, purpose (plan/draft/verify/embed), model, tokens, latency, outcome, error code.

Rules enforced by the database

#	Rule	Enforcement
1	One unfinished attempt per user and subject	Partial unique index on (user_id, subject_id) WHERE status IN (GENERATING, READY, IN_PROGRESS, AWAITING_SUBMISSION). Conflict returns 409 ATTEMPT_EXISTS with the existing attempt ID.
2	One active generation job per subject	Partial unique index on subject_id WHERE state IN (QUEUED, RUNNING, STALLED, CANCELLING).
3	Four choices, exactly one correct	Unique (question_id, position) with CHECK position BETWEEN 1 AND 4; partial unique index on question_id WHERE is_correct. A question is inserted only in a transaction that writes all four choices and verifies the count.
4	Exactly 50 questions before READY	Unique (attempt_id, position) with CHECK position BETWEEN 1 AND 50. READY is a single conditional UPDATE that succeeds only when 50 rows exist and all questions are ACTIVE.
5	One answer per attempt question	Unique (attempt_question_id).
6	Idempotent, atomic submit	Unique (attempt_id) on attempt_result; status change via UPDATE ... WHERE status = 'AWAITING_SUBMISSION' and a row-count check.
7	Immutable questions and results	BEFORE UPDATE triggers reject changes to question and choice content and to any attempt_result row. Deletion happens only through subject-deletion cascades.
8	Monotonic revisions	Writes use UPDATE ... WHERE revision = :expected; the server increments. Client clocks are never used.
9	Stale session rejected	Every mutating call carries sessionVersion; UPDATE ... WHERE session_version = :v.
10	Valid citations	FK to document; CHECK page > 0; page validated against document page count at insert.
11	Report ownership; one report per question per attempt	FKs plus unique (reporter, attempt, question); ownership check in the service.
12	Keys never leave in exam responses	Exam DTO classes contain no key, explanation or citation fields; a contract test fails if any serialized exam response contains them.

6. Question generation and validation



Generation pipeline – every accepted question is checkpointed before the next is drafted

6.1 Readiness gate

A job starts only when the subject has at least one READY document and meets the content threshold (App. A). Otherwise the API returns INSUFFICIENT_CONTENT with the actual counts (eligible chunks, pages), so the UI can say what is missing instead of failing later.

6.2 Planning and coverage

- **Snapshot** the document IDs and index versions into GenerationJobDocument. Documents added later are ignored by this job.
- **Topics**: cluster eligible chunks by chapter/section and embedding similarity, name the clusters, and store them as SubjectTopic. Topics are stable afterwards; new ones are appended.
- **Slots**: allocate the target count across topics in proportion to eligible content, capped at 3 per topic (raised only as far as needed when a subject has fewer than 17 topics). Each slot gets 1-3 adjacent seed chunks.
- **Answer-position balance** is done by code, not by the model: when an attempt is created, the correct choice is assigned a balanced position (each of A-D appears 10-15 times in 50) and the distractors are shuffled around it. The order is persisted once.

6.3 Drafting

One slot at a time. Input: seed chunks, topic, and the objectives already accepted. Output: strict JSON against a schema (five-sentence vignette, lead-in, four choices, one key, objective, explanations for the key and every distractor, topic, citation with quote). Free-text fallback is not allowed; a parse failure counts as a rejected draft. Source text is passed as delimited data, never as instructions.

6.4 Validation layers

Layer	What it checks
A. Deterministic (no AI)	Schema valid. Vignette is exactly five sentences. Lead-in present. Exactly four distinct choices, one key, no "all/none of the above", choice lengths not wildly uneven. All four explanations present. Topic is in SubjectTopic. Citation document is in the job snapshot, page is within the page count, and the quote is found on that page by normalized fuzzy match.
B. Independent verifier (AI)	A separate call, preferably a different model, receives only the cited text, vignette, lead-in and choices. It never sees the key or the explanations. For each choice it returns SUPPORTED, CONTRADICTED or NOT_ADDRESSED plus the supporting quote, and flags NEEDS_OUTSIDE_KNOWLEDGE or AMBIGUOUS. Accept only if exactly the keyed choice is SUPPORTED and no flag is raised. Any disagreement rejects the draft. The explanations are checked against the source in the same call.
C. Duplicate check	Cosine similarity of the vignette-plus-lead-in embedding against accepted questions in this job and in the active bank. Reject at or above the threshold; also reject the same topic with the same normalized educational objective.
D. Full-set validation	After the target is reached: count is 50, IDs unique, every citation still valid and its document still existing and in the snapshot, topic cap respected, minimum distinct topics met, no pair above the similarity threshold, all questions ACTIVE. Failures replace only the offending slots, within a bounded budget.

Honest limit

Validation reduces medical errors; it cannot eliminate them. The verifier only proves that the answer is supported by the uploaded text, not that the text is clinically right. That is why the report path (section 8), the study-aid disclaimer (section 10) and the M2 manual-review gate exist.

6.5 Limits

Every limit has a default in Appendix A and lives in configuration: attempts per slot, total AI calls per job, per-call timeout and retries with exponential backoff and jitter (honoring Retry-After), job wall-clock timeout, and the replacement budget. Exceeding any limit ends the job with an explicit reason, never a silent loop.

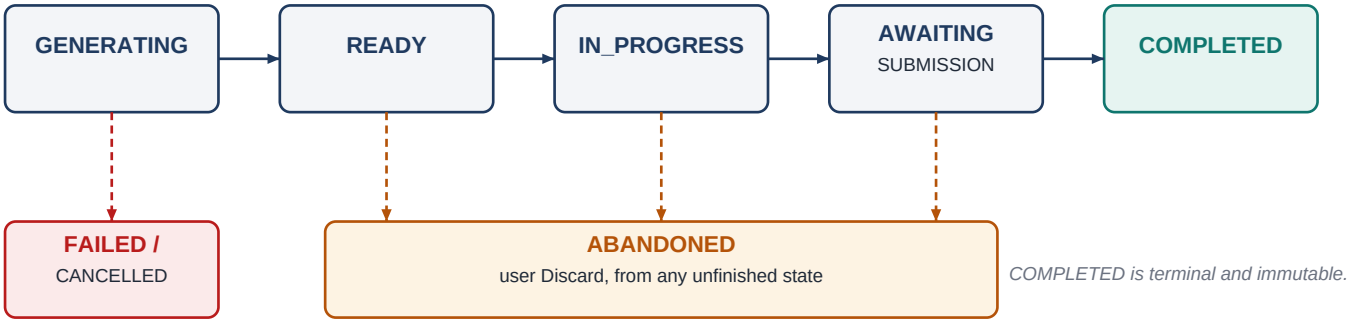
6.6 Job state machine, recovery and cancellation

- **States**: QUEUED → RUNNING → COMPLETED. RUNNING → STALLED (watchdog) → RUNNING (auto-recovery, limited) or FAILED. Any active state → CANCELLING → CANCELLED.
- **Failure reasons** (shown in the UI): INSUFFICIENT_CONTENT, PROVIDER_UNAVAILABLE, BUDGET_EXHAUSTED, VALIDATION_EXHAUSTED, SOURCE_CHANGED, TIMEOUT, INTERNAL.
- **Checkpoint**: each accepted question is inserted with its choices and citations and the job counter is incremented in one transaction. Recovery replans only the remaining slots. Progress counts accepted, validated questions, never attempted AI calls.
- **Heartbeat and watchdog**: the worker writes heartbeatAt; a job with no heartbeat for the stall window is marked STALLED and resumed. On boot the worker scans for RUNNING jobs left by a previous process. On graceful shutdown it finishes the current call and returns the job to QUEUED.
- **Cancel** requires confirmation and sets CANCELLING in a transaction. The worker checks the flag before every AI call and every persist, and persists with a guard (job state must be RUNNING), so a late result cannot be published. Content from the cancelled job is discarded.
- **Content changes mid-job**: new PDFs are ignored (snapshot). A deleted PDF causes questions citing it to be rejected and replaced; losing too many slots fails the job with SOURCE_CHANGED. A deleted subject cancels the job. Reprocessing does not disturb the job because the old index version stays pinned.

6.7 Building an attempt from the bank

Eligible = ACTIVE questions whose citations are not all unavailable. Selection prefers questions the user has not seen (a question counts as seen once it appeared in an attempt that reached IN_PROGRESS), then least recently seen, while respecting the topic cap. If fewer than 50 eligible questions exist, a job generates the shortfall. If enough eligible questions exist but fewer than 50 are unseen, the user chooses explicitly: include repeats, or generate new ones (uses AI budget). Selection and attempt creation run in one transaction.

7. Exam state machine



Attempt lifecycle – only COMPLETED is immutable; every unfinished state can be discarded

State	Transitions and UI behavior
GENERATING	Background progress (0-50) that survives navigation. Cancel with confirmation, or recover a failed job.
READY	Show Start Test; no automatic navigation. Discard available.
IN_PROGRESS	Forward-only Answer & Continue or Skip. Back exits to the dashboard; progress persists.
AWAITING_SUBMISSION	All 50 answered; editable neutral navigator; explicit Submit.

State	Transitions and UI behavior
COMPLETED	Immutable grade, Results, View Answers, Retry, Done.
FAILED / CANCELLED	Meaningful status and reason; safe retry or new generation.
ABANDONED	Result of Discard from any unfinished state. Frees the one-unfinished-attempt slot.

Forward-only navigation (server-computed)

First pass traverses positions 1-50 in the persisted order. Skip marks a question SKIPPED. The server computes the next question with one deterministic rule: the smallest position greater than the current one whose status is UNSEEN or SKIPPED; if none, the smallest position with that status; if all are answered the attempt moves to AWAITING_SUBMISSION. The client never decides, so refresh, resume and revisit all behave identically. No Previous button and no number navigator during the active exam.

- **Answer immutability during IN_PROGRESS**: an answered question cannot change. An identical replay returns 200 with no change; a different value returns 409 REVISION_CONFLICT. Editing is allowed only in AWAITING_SUBMISSION.
- **Pre-submission review**: all 50 numbers neutral, only the selected one highlighted. Jump, Previous/Next, edit with autosave. No keys, explanations, citations or correctness indicators before submission.

v1 connectivity and session ownership

- **Online-only autosave.** Each Answer or Skip is saved to the server with an idempotency key. When offline, controls are disabled and a banner reads "Offline – answers not saved". Failed saves retry with backoff; a refresh restores state from the server. Submission is blocked while any save is in flight.
- **Latest session wins.** Opening the exam increments sessionVersion. A mutating call from an older tab returns 409 SESSION_SUPERSEDED; that tab becomes read-only with a "Use this window" action. Nothing is silently merged.

Submission

One endpoint with an idempotency key. In a single transaction: lock the attempt row; if already COMPLETED return the existing result; require status AWAITING_SUBMISSION and all 50 answers; grade on the server from the stored keys; insert the result snapshot; set COMPLETED. Twenty concurrent submits produce exactly one result.

Stuck-state handling

- Discard (any unfinished state, including FAILED) moves the attempt to ABANDONED after confirmation.
- The dashboard shows attempt age and highlights unfinished attempts older than 14 days; there is no silent auto-expiry.
- Watchdog and boot recovery (section 6.6) cover stuck GENERATING attempts; the partial unique index guarantees the dashboard can never show two unfinished attempts for one subject.

8. Results, review and reports

Each question grades 1 or 0; the score is /50 with percentage and correct/incorrect counts. No pass/fail label and no exam timer. Results include topic breakdowns (using the controlled topic list) and Retry, View Answers, Done. Post-exam review shows the original shuffled choices, selected and correct answers, explanations, citations, and All/Correct/Incorrect filters combinable with a topic filter.

Report lifecycle

Event	Effect
User reports a question (answer key, ambiguity, citation, explanation, other)	Report is OPEN. The score is never changed automatically.
Reason is ANSWER_KEY or CITATION	Question is QUARANTINED immediately: excluded from new attempts. Past attempts keep it, marked "Under review".
Owner reviews the Reports screen	ACCEPTED → question RETIRED permanently. DISMISSED → question returns to ACTIVE if this report was the only reason it was quarantined.
Question later retired	History shows a notice on that question; the stored grade stays as recorded (D10).

9. API and trust boundaries

API capability	Security / consistency rule
Subject and PDF operations	Authenticated ownership; reuse existing routes and storage workflow; upload limits and magic-byte checks.
Generation start / status / cancel	Idempotency key; one active job; readiness gate; budget check; job ownership; race-safe transitions; status polling with backoff.
Attempt create / discard	Single transaction; partial unique index backs the rule; Discard only on the owner's unfinished attempts.
Exam retrieval	Exam-safe DTO: no keys, correctness, explanations or citations; opaque choice IDs.
Answer / skip	sessionVersion check, revision check, status rule (immutable during IN_PROGRESS), server-computed next question.
Submit	Idempotent; server-only grading; complete-answer check; atomic transaction.
Results / history / review	Only COMPLETED attempts; immutable answer mapping and citation metadata.
Question reports	Authenticated reporter; attempt and question ownership; one report per question per attempt.

Stable error codes the client can rely on: ATTEMPT_EXISTS, JOB_ACTIVE, INSUFFICIENT_CONTENT, BUDGET_EXHAUSTED, SESSION_SUPERSEDED, REVISION_CONFLICT, NOT_ALL_ANSWERED, SOURCE_UNAVAILABLE.

Security requirements

- Uploaded PDF text and model outputs are untrusted. Source text goes to the model as delimited data; the model has no tools; output must match the schema or it is discarded; secrets never appear in prompts.
- Model text is never rendered as raw HTML. Render as escaped text or sanitized markdown (stored XSS via generated questions is a real bug class).
- PDFs are served through signed URLs with a short TTL, issued per request after the ownership check. No public R2 links.
- API keys and provider routing stay on the server. DTO whitelisting and validation everywhere; audit-friendly logging of destructive actions.
- **[PUBLIC]** per-user and per-IP throttles on upload, generation start and report endpoints.
- Additive, reviewed migrations with a rollback note each. Never reset the database; never alter unrelated behavior.

10. Operations, cost and rollout

Deployment and reliability

Use the existing topology. The worker runs in the API process and recovers on boot. Closing the machine or stopping the API pauses generation; the UI then shows "Paused – server not running" when the heartbeat is stale, and the job resumes from its checkpoint when the API returns.

Cost and abuse controls

- One active job at a time; per-job caps on drafts, verifications and total AI calls.
- Configurable daily AI-call budget; exhausting it fails new jobs with BUDGET_EXHAUSTED and leaves running jobs to finish their current slot.
- Circuit breaker around OmniRoute: opens after consecutive failures, waits through a cooldown, then probes. While open, jobs wait inside their timeout rather than burning calls.
- Every AI call is recorded in ai_call_log, so cost per accepted question is measurable.

Observability

Structured logs with jobId and attemptId. Track: job success and failure by reason, rejection rate per validation layer, AI calls per accepted question, stalled jobs, provider 429 and error rates, and processing lag. A health endpoint reports these; investigate any stalled job older than 10 minutes or a rejection rate that jumps after a prompt or model change.

Backup, restore and data hygiene

- Scheduled database dump (daily or weekly) stored off the machine; restore drill run once before release. Embeddings are re-derivable from R2 but costly, so chunks are included in the dump. Do not assume the hosting plan provides backups.
- Reconciler for orphaned R2 objects and missing objects (section 4).

Rollout and compatibility

- Feature flag on server and client (`MOCK_TEST_ENABLED`). Off hides navigation and returns 404 on Mock Test routes. Migrations are additive, so the off state is harmless to chat and Subjects.
- Migration check on real data: backfill document status, page counts and index version for existing PDFs; compare row counts before and after; verify existing chat still retrieves the same chunks.
- Mobile and accessibility: usable at 360 px width, full keyboard operation, visible focus, screen-reader labels on choices.

Product and legal

Show a study-aid notice on generation and results screens: questions are AI-generated from your own PDFs, are not medical advice, and may contain errors; use Report. State that the user is responsible for the right to use the uploaded material and that it is for personal study.

11. Testing strategy

Layer	What is tested
Unit	Traversal rule, grading, balanced answer-position assignment, sentence counter, quote fuzzy match, bank selection.
Database constraints	Each rule in section 5 has a test that tries to violate it, including concurrency: 20 parallel attempt creations, 20 parallel submits, racing answer writes, stale-session writes.
API contract	Exam-safe responses never contain isCorrect, explanations or citations (snapshot test). Error codes match section 9.
Job chaos	Kill the process mid-job: resumes with no duplicate or partial questions and ends at exactly 50. Cancel during an AI call: nothing is published. Provider 429 storm: breaker opens and the job ends cleanly. PDF deleted mid-job.
Golden set	3-5 PDFs (text, scanned, mixed) with fixed expectations: citation validity 100%, no near-duplicate pairs, topic cap respected, acceptance rate within a band. Run on every prompt, model or validator change.
Manual quality review	Read a 50-question sample before release (M2, M8) and 20 after any significant prompt or model change.
Injection and XSS	PDF containing "ignore previous instructions" and markup: schema unchanged, nothing leaked, nothing rendered as HTML.
End-to-end	Full exam flow on desktop and a 360 px viewport, keyboard-only, refresh at every state.
Regression	The existing backend suite stays green at every milestone; existing chat, upload and auth flows re-verified at M8.

Exit rule: a milestone closes only when its own tests pass and the existing baseline is green.

12. Phased delivery and acceptance

Milestones are numbered M0-M9 so they do not collide with the existing backend phase tags.

Milestone	Focus	Exit criterion
M0	Decisions	D1-D11 and Appendix A confirmed or changed in writing; this document frozen as v2.1.
M1	Repository audit	Existing services, schema, tests and risks documented; deployment assumption confirmed; baseline verified green.
M2	Generation spike (script only)	3-5 real PDFs through plan, draft, layers A-D; manual review of at least 50 questions. Go if citation validity is 100% and at most 2 of 50 have a wrong or unsupported key. Measure acceptance rate and AI calls per accepted question; calibrate the similarity threshold and retry limits. No schema, no UI.
M3	Data and API foundation	Additive migrations with raw-SQL constraints, feature flag, bank model; every section 5 rule has a violating test; concurrency tests pass.

Milestone	Focus	Exit criterion
M4	Exam flow on fixed questions	A seeded 50-question fixture runs the whole flow (create attempt, traversal, skip and revisit, autosave, latest-session-wins, submit, grade, results) with no AI involved.
M5	Real generation	Worker, queue, state machine, readiness gate, bank selection, cancel and recovery, budget and breaker; job chaos tests pass.
M6	Citations, review and reports	Authenticated viewer, Source Unavailable, report lifecycle, filters, topic analytics verified.
M7	Hardening	Upload hygiene, reconciler, backup and restore drill, observability and health endpoint, mobile and accessibility, injection and XSS tests.
M8	Regression and rollout	Frontend and backend builds, integration, existing flows pass; migration check on real data; flag enabled for personal use; a week of real use with a fix list closed.
M9 (optional)	v1.1	Offline sync and explicit takeover (section 13).

If M2 fails (cut line)

Do not build further. Options, in order: tighten source windows (one chunk per question), add a "preview and approve" step before a set is published, restrict generation to documents with clean text, or revise the 50-question rule. Decide before M3, not after the exam UI exists.

13. Deferred to v1.1

Offline sync

- Local queue in IndexedDB keyed by user, attempt and sessionVersion. Operations carry an idempotency key and the base revision.
- Progress allowed only through already-loaded questions. Sync on reconnect, in order. Submission blocked until the queue is empty.
- If the session was superseded, local operations are discarded with a message stating how many answers were lost. Logout clears the store. If storage quota fails, stop progression and say so.

Explicit takeover

Replace latest-wins with a lease: server-timestamped heartbeat (about 30 s) and TTL (about 90 s), a takeover prompt when another session is live, and revocation of the old session's authority.

14. Final acceptance checklist

- Pediatrics created in Mock Test appears in Subjects; two attached PDFs show "2 PDFs", never "50 items".
- Only processed, PDF-grounded material is used; every question passes validation layers A to D; every citation resolves to a real page.
- A set covers the subject: topic cap respected, minimum distinct topics met, answer positions balanced.
- Question and choice order are stable across refresh and resume.
- Skip, revisit and pre-submission editing follow the rules; answered questions cannot change during IN_PROGRESS.
- Concurrent attempt creation, answers and submits cannot create duplicate or inconsistent state.
- Stale tabs cannot overwrite progress; offline shows a clear non-saving state.
- A stuck attempt or job can always be discarded, cancelled or recovered; nothing blocks a subject permanently.
- Killing the worker mid-generation never produces partial or duplicate questions.
- Results, citations, reports and history remain consistent when PDFs are deleted; reported questions leave new attempts.
- Spend is bounded: job caps, daily budget and breaker verified.
- Backup restore drill completed; reconciler dry-run clean.
- Existing Subjects, chat, authentication and upload workflows pass regression with the flag on and off.

Implementation note

Before coding, reconcile this blueprint with the real repository. Where the current implementation differs, propose the smallest safe migration path and verify each milestone before proceeding.

Appendix A. Parameter defaults

Starting values. Calibrate in M2 and keep them in configuration, not code.

Parameter	Default	Notes
Minimum eligible chunks to generate	60	Readiness gate (D6).
Minimum distinct pages with eligible text	40	Readiness gate (D6).
Max questions per topic in one set	3	Raised only to ceil(50 / topic count) for small subjects.
Minimum distinct topics per set	17	Follows from the cap of 3.
Correct-answer position balance	10-15 per letter in 50	Assigned by code at attempt creation.
Draft attempts per slot	3	Then the slot is reassigned.
Total AI calls per job (hard cap)	400	Counts drafts, verifications and embeddings.
Replacement budget at full-set validation	5 slots	Beyond this the job fails VALIDATION_EXHAUSTED.
Duplicate similarity threshold	0.90 cosine	Calibrate in M2.
Citation quote fuzzy-match threshold	0.90	On normalized text of the cited page.
Job wall-clock timeout	45 min	Failure reason TIMEOUT.
Heartbeat / stalled window	30 s / 120 s	Watchdog marks STALLED.
Auto-recoveries per job	3	Then FAILED.
AI call timeout / retries	60 s / 3	Backoff 2 s, 4 s, 8 s plus jitter; honor Retry-After.
Circuit breaker	Open after 5 consecutive failures; 60 s cooldown	Then a single probe call.
Daily AI-call budget	1,500	Roughly five full jobs; configurable.
Max PDF size / pages	50 MB / 800 pages	Magic bytes %PDF- and MIME checked.
OCR cap per document	200 pages	Over 30% unreadable pages fails with OCR_QUALITY.
Per-document processing timeout	15 min	Failure reason TIMEOUT.
Minimum chunk length for questions	200 characters	Shorter chunks are chat-only.
Signed URL TTL	5 min	Issued per request after ownership check.
Stale-attempt highlight	14 days	Display only; no auto-expiry.
Orphan R2 object grace period	24 h	Dry-run report before deletion.
Status polling interval	3 s with backoff	Generation progress.